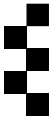
Evaluation of IR System



University of Essex

A Classification System

Consider a two-class classification problem

Gender Recognition system



University of Essex

Evaluation of a System: why it needed?

- Concerned (positive) class: Male
- Negative Class: Female
- Testing set contains 150 male and 30 female **(total 180 images)**

		Actual Class		Total Predicted
		Male	Female	
Predicted Class	Male	135	6	141
	Female	15	24	39
Total Actual		150	30	180

Accuracy of a Classification System (1)

- Testing set contains 150 male and 30 female **(total 180 images)**

$$Accuray = \frac{\text{True Male + True Female}}{\text{Total Images}} \times 100 = \frac{135+24}{180} \times 100 = 88.3\%$$

		Actual Class		Total Predicted
		Male	**Female**	
Predicted Class	**Male**	**TP=135 (True Positive)**	**FP=6 (False Positive)**	TP+FP=141 (Predicted Positive)
	Female	**FN=15 (False Negative)**	**TN=24 (True negative)**	TN+FN=39 (Predicted Negative)
Total Actual		TP+FN=150 (Actual Positive)	TN+FP=30 (Actual Negative)	**180**

Accuracy of a Classification System (2)

Testing set contains 150 male and 30 female **(total 180 images)**

$$Accuray = \frac{\text{True Male + True Female}}{\text{Total Images}} \times 100 = \frac{135+24}{180} \times 100 = 88.3\%$$

$$Accuray = \frac{TP + TN}{TP+FN+TN+FP} \times 100 = \frac{135+24}{180} \times 100 = 88.3\%$$

		Actual Class		Total Predicted
		Male	**Female**	
Predicted Class	**Male**	**TP=135 (True Positive)**	**FP=6 (False Positive)**	TP+FP=141 (Predicted Positive)
	Female	**FN=15 (False Negative)**	**TN=24 (True negative)**	TN+FN=39 (Predicted Negative)
Total Actual		TP+FN=150 (Actual Positive)	TN+FP=30 (Actual Negative)	**180**

Accuracy: is it sufficient?

$$Accuray = \frac{TP + TN}{TP+FN+TN+FP} \times 100$$

So we have defined the accuracy

Is it sufficient to evaluate a system

or

Do we need more measures?

Let's see!

University of Essex

Accuracy as a Poor Measure

Testing set contains 150 male and 30 female **(total 180 images)**

$$Accuray = \frac{TP + TN}{TP+FN+TN+FP} \times 100 = \frac{135+0}{180} \times 100 = 75\%$$

- This accuracy (75%) is useless.
- The system simply recognizing every class as a positive class (male).
- **In fact, very poor system.**

		Actual Class		**Total Predicted**
		Male	**Female**	
Predicted Class	**Male**	**TP=135 (True Positive)**	**FP=30 (False Positive)**	TP+FP=165 (Predicted Positive)
	Female	**FN=15 (False Negative)**	**TN=0 (True negative)**	TN+FN=15 (Predicted Negative)
Total Actual		TP+FN=150 (Actual Positive)	TN+FP=30 (Actual Negative)	**180**

Recognition Rate of Positive Class

Testing set contains 150 male and 30 female **(total 180 images)**

$$\text{Recog. of Pos. Class} = \frac{\text{True Male}}{\text{Total Actual Male}} \times 100 = \frac{135}{150} \times 100 = 90\%$$

$$\textit{sensitivity} = \frac{\text{TP}}{\text{TP + FN}} \times 100 = \frac{135}{150} \times 100 = 90\%$$

		Actual Class		**Total Predicted**
		Male	**Female**	
Predicted Class	**Male**	**TP=135 (True Positive)**	**FP=30 (False Positive)**	TP+FP=165 (Predicted Positive)
	Female	**FN=15 (False Negative)**	**TN=0 (True negative)**	TN+FN=15 (Predicted Negative)
Total Actual		TP+FN=150 (Actual Positive)	TN+FP=30 (Actual Negative)	**180**

Recognition Rate of Negative Class

Testing set contains 150 male and 30 female **(total 180 images)**

$$\text{Recog. of Neg. Class} = \frac{\text{True Female}}{\text{Total Actual Female}} \times 100 = \frac{0}{30} \times 100 = 0\%$$

$$\textit{specificity} = \frac{\text{TN}}{\text{TN + FP}} \times 100 = \frac{0}{30} \times 100 = 0\%$$

		Actual Class		**Total Predicted**
		Male	**Female**	
Predicted Class	**Male**	**TP=135 (True Positive)**	**FP=30 (False Positive)**	TP+FP=165 (Predicted Positive)
	Female	**FN=15 (False Negative)**	**TN=0 (True negative)**	TN+FN=15 (Predicted Negative)
Total Actual		TP+FN=150 (Actual Positive)	TN+FP=30 (Actual Negative)	**180**

Evaluation of a Classification System

Testing set contains 150 male and 30 female **(total 180 images)**

$$sensitivity = \frac{TP}{TP + FN} \times 100 = \frac{135}{150} \times 100 = 90\%$$

$$specificity = \frac{TN}{TN + FP} \times 100 = \frac{0}{30} \times 100 = 0\%$$

$$Accuray = \frac{TP + TN}{TP+FN+TN+FP} \times 100 = \frac{135+0}{180} \times 100 = 75\%$$

		Actual Class		Total Predicted
		Male	**Female**	
Predicted Class	**Male**	**TP=135 (True Positive)**	**FP=30 (False Positive)**	TP+FP=165 (Predicted Positive)
	Female	**FN=15 (False Negative)**	**TN=0 (True negative)**	TN+FN=15 (Predicted Negative)
Total Actual		TP+FN=150 (Actual Positive)	TN+FP=30 (Actual Negative)	**180**

More Performance Measures: Recall

Testing set contains 150 male and 30 female **(total 180 images)**

$$sensitivity = \frac{\text{TP}}{\text{TP + FN}} \times 100 = \frac{135}{150} \times 100 = 90\%$$

sensitivity also called Recall

		Actual Class		Total Predicted
		Male	**Female**	
Predicted Class	**Male**	**TP=135 (True Positive)**	**FP=6 (False Positive)**	TP+FP=141 (Predicted Positive)
	Female	**FN=15 (False Negative)**	**TN=24 (True negative)**	TN+FN=39 (Predicted Negative)
Total Actual		TP+FN=150 (Actual Positive)	TN+FP=30 (Actual Negative)	**180**

More Performance Measures: Precision

Testing set contains 150 male and 30 female **(total 180 images)**

$$precision = \frac{\text{True Male}}{\text{Total Predicted Male}} \times 100 = \frac{135}{141} \times 100 = 95.7\%$$

$$precision = \frac{TP}{TP + FP} \times 100 = \frac{135}{141} \times 100 = 95.7\%$$

		Actual Class		Total Predicted
		Male	**Female**	
Predicted Class	**Male**	**TP=135 (True Positive)**	**FP=6 (False Positive)**	TP+FP=141 (Predicted Positive)
	Female	**FN=15 (False Negative)**	**TN=24 (True negative)**	TN+FN=39 (Predicted Negative)
Total Actual		TP+FN=150 (Actual Positive)	TN+FP=30 (Actual Negative)	**180**

Difference between Recall and Precision

Both measures provide the information about positive class (concerned class).

$$recall = \frac{\text{True Male}}{\text{Total Actual Male}} \times 100 = \frac{135}{150} \times 100 = 90\%$$

$$recall = \frac{\text{TP}}{\text{TP + FN}} \times 100 = \frac{135}{150} \times 100 = 90\%$$

Fraction of accurately detected positive samples from actual positive samples.

$$precision = \frac{\text{True Male}}{\text{Total Predicted Male}} \times 100 = \frac{135}{141} \times 100 = 95.7\%$$

$$precision = \frac{\text{TP}}{\text{TP + FP}} \times 100 = \frac{135}{141} \times 100 = 95.7\%$$

Fraction of accurately detected positive samples from predicted positive samples.

More Performance Measures: F1-score

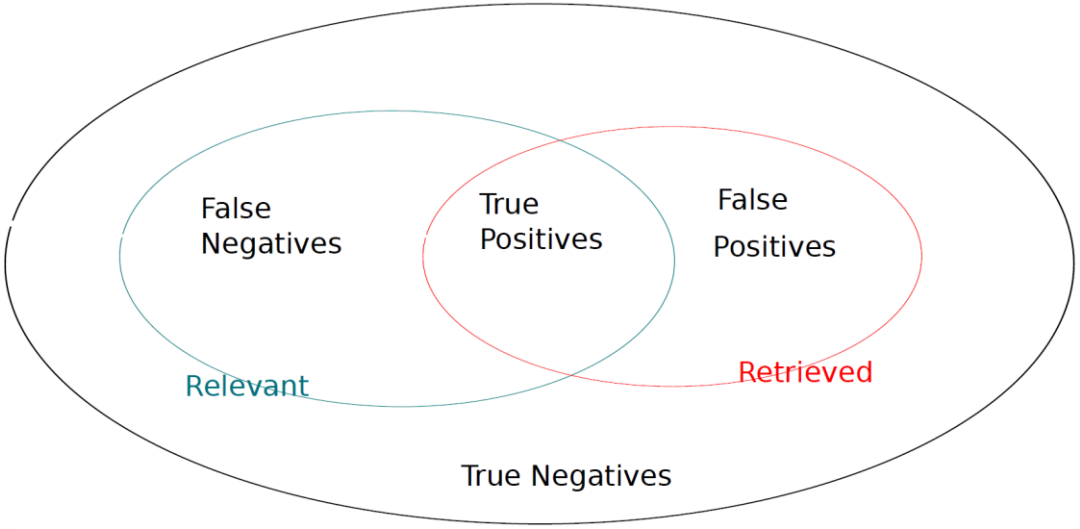
Testing set contains 150 male and 30 female **(total 180 images)**

$$F1 = 2 \times \frac{precision \times recall}{precision + recall} \times 100 = 2 \times \frac{90 \times 95.7}{90 + 95.7} \times 100 = 92.7\%$$

$$F1 = \frac{2TP}{2TP + FP + FN} \times 100 = \frac{270}{270 + 6 + 15} \times 100 = 92.7\%$$

		Actual Class		**Total Predicted**
		Male	**Female**	
Predicted Class	**Male**	**TP=135 (True Positive)**	**FP=6 (False Positive)**	TP+FP=141 (Predicted Positive)
	Female	**FN=15 (False Negative)**	**TN=24 (True negative)**	TN+FN=39 (Predicted Negative)
Total Actual		TP+FN=150 (Actual Positive)	TN+FP=30 (Actual Negative)	**180**

Evaluation of IR System



		TRUTH	
		Relevant	**Not Relevant**
What system think	**Retrieved**	**TP (True Positive)**	**FP (False Positive)**
	Not Relevant	**FN (False Negative)**	**TN (True negative)**

Evaluation of IR System: Precision

$$recall = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

$$recall = \frac{\text{number of relevant documents which are retreived}}{\text{number of relevant documents}}$$

		TRUTH	
		Relevant	Not Relevant
What system think	Retrieved	TP (True Positive)	FP (False Positive)
	Not Relevant	FN (False Negative)	TN (True negative)

Evaluation of IR System: Precision

$$precision = \frac{TP}{TP + FP}$$

$$precision = \frac{\text{number of relevant documents which are retreived}}{\text{number of retreived documents}}$$

		TRUTH	
		Relevant	Not Relevant
What system think	Retrieved	TP (True Positive)	FP (False Positive)
	Not Relevant	FN (False Negative)	TN (True negative)

Exercise 1

Calculate **Precision, Recall, and F1.**

		TRUTH	
		Relevant	**Not Relevant**
What system think	**Retrieved**	**TP = 20 (True Positive)**	**FP = 40 (False Positive)**
	Not Relevant	**FN = 60 (False Negative)**	**TN = 1,000,000 (True negative)**

Precision and Recall Trade-off

- You can increase recall by returning more docs.
- Recall is a non-decreasing function of the number of docs retrieved.
- A system that returns all docs has 100% recall!
- The converse is also true (usually): It's easy to get high precision for very low recall.

Exercise 2

Calculate **Precision, and Recall.**

		TRUTH	
		Relevant	**Not Relevant**
What system think	**Retrieved**	**TP = 80 (True Positive)**	**FP = 800 (False Positive)**
	Not Relevant	**FN = 0 (False Negative)**	**TN = 999,240 (True negative)**

Exercise 3

Calculate **Precision, and Recall.**

		TRUTH	
		Relevant	**Not Relevant**
What system think	**Retrieved**	**TP = 5 (True Positive)**	**FP = 10 (False Positive)**
	Not Relevant	**FN = 75 (False Negative)**	**TN = 1,000,030 (True negative)**

Suggested Readings

- Book: *An Introduction to Information Retrieval*
  - **Chapter 6:** Section 6.2 and 6.3
  - **Chapter 8:** Section 8.3